

Experiment 5

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1 Aim

In this experiment we aim to study 555 Timer IC as an Astable multivibrator with different frequency.

2 Theory

The 8-pin 555 timer is one of the most useful ICs ever made and it is used in many projects. It is one of the most widely used integrated circuits which can be used to construct astable and monostable multivibrators. It can also be used in many other digital and analog circuits. The diagram below shows the internal structure of IC 555 timer.

2.1 Internal Structure of 555 Timer

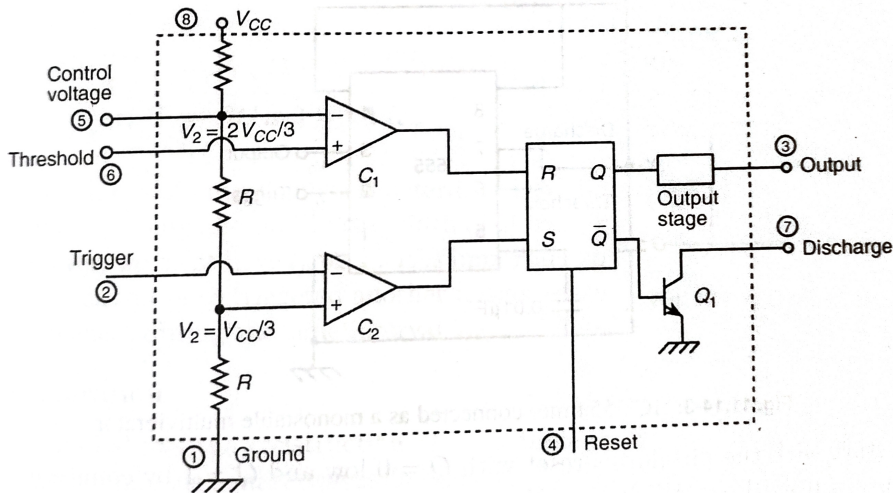


Figure 1: Internal Structure of IC 555

A 555 Timer IC consists (Fig 1) of two comparators, one SR flip-flop, a discharge transistor (Q_1), a resistive voltage divider string and an output amplifier stage. Standard 555 and 556 ICs create a significant *glitch* on the supply when their output changes state. This is rarely a problem in simple circuits with no other ICs, but in more complex circuits a *smoothing capacitor* (eg $100\ \mu F$) should be connected across the $+V_S$ (or V_{CC}) and $0V$ supply near the 555 or 556.

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The entire circuit is usually housed in an eight-pin package (shown in Fig 2). When used with $V_{CC} = +5\text{ V}$, its operation becomes compatible with TTL¹ and CMOS logic circuits.

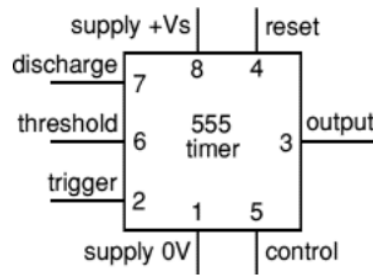


Figure 2: Pin Diagram of IC 555

The working of the IC timer can be understood easily by analyzing how the comparator outputs affect the SR flip-flop. The **threshold** and **trigger** inputs control the outputs of the comparators C_1 and C_2 . The inverting input of C_1 is at $\frac{2}{3}V_{CC}$ and the non-inverting input of C_2 is at $\frac{1}{3}V_{CC}$. Therefore, if the threshold input is greater than $\frac{2}{3}V_{CC}$, the output of C_1 goes high; if the trigger input is less than $\frac{1}{3}V_{CC}$, the output of C_2 goes high.

1. When the outputs of both the comparators are low ($R = S = 0$), the output of the flip-flop remains unchanged.
2. When the output of C_1 is high (1) and that of C_2 is low (0), then the output Q of the flip-flop goes low.
3. When the output of C_1 is low (0) and that of C_2 is high (1), then the output Q of the flip-flop goes high.

2.2 Astable Operation

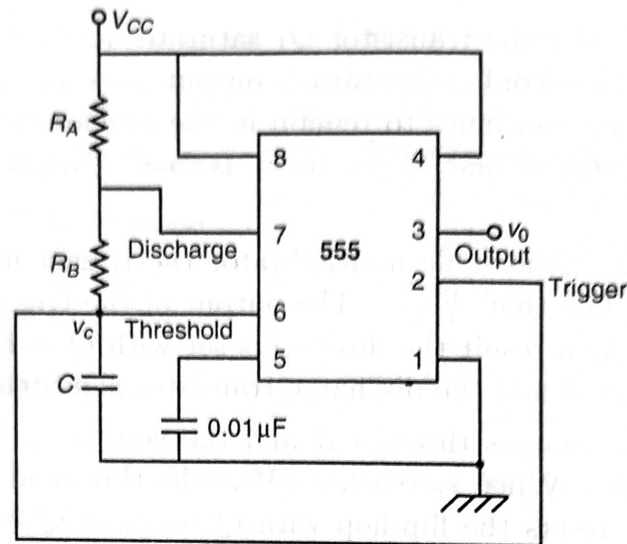


Figure 3: IC Timer 555 connected as an *astable multivibrator*

Fig 3 shows the construction of an astable multivibrator using IC 555. It can produce continuous square wave frequency of which is determined by the values of R_A , R_B and capacitor C . Here the threshold and trigger inputs are connected together.

In the astable configuration, the 555 timer does not remain stable in any state. It continuously switches between high and low output levels.

¹ Transistor Transistor Logic [https://www.wikiwand.com/en/articles/Transistor%E2%80%93transistor_logic]

To start with let us assume that the capacitor C is in discharged state. Then the trigger input is less than $\frac{1}{3}V_{CC}$. As a result, the trigger comparator goes high ($S = 1$) and sets the flip-flop with $Q = 1$ and $\overline{Q} = 0$. The output of the timer is high now and the discharge transistor is switched off. The capacitor now charges through R_A and R_B towards V_{CC} . When voltage v_c across the capacitor just exceeds $\frac{2}{3}V_{CC}$, the threshold comparator goes high and resets the flip-flop with $Q = 0$ and $\overline{Q} = 1$. The output of the timer goes low and the discharge transistor Q_1 is switched on. The capacitor now starts discharging through R_B and the transistor Q_1 . When the capacitor voltage v_c just goes below $\frac{1}{3}V_{CC}$ the trigger comparator again goes high and sets the flip-flop making timer output high. The cycle repeats and the output of the timer repeatedly goes high and low producing continuous square waves.

Note that the capacitor periodically charges up to a maximum value $\frac{2}{3}V_{CC}$ and discharges to a minimum value $\frac{1}{3}V_{CC}$.

The waveforms of the **timer output** V_{out} and the voltage v_c across the capacitor C are shown in Fig 4 . The period of the output waveform is given by the sum of the **mark time** t_m required by the capacitor to charge from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$ through R_A , R_B and the **space time** t_s required to discharge from $\frac{2}{3}V_{CC}$ to $\frac{1}{3}V_{CC}$ through R_B .

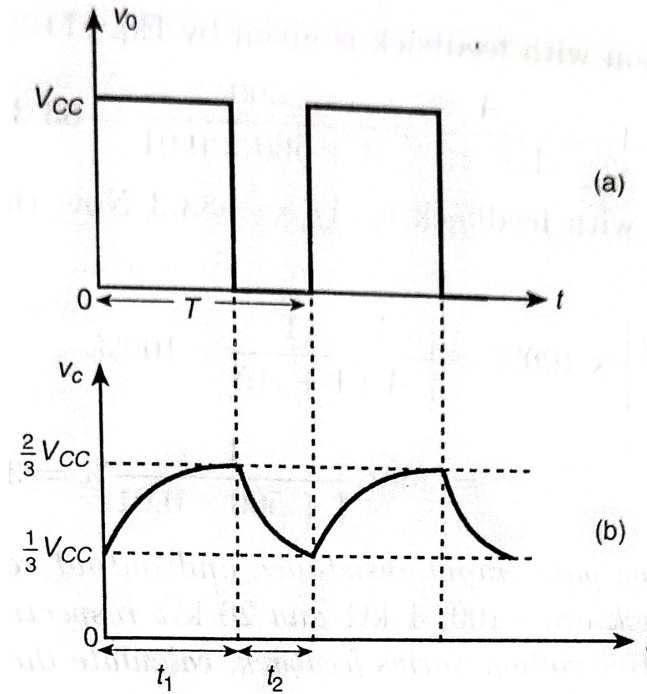


Figure 4: Waveforms for (a) V_{out} and (b) v_c

Calculation of charging/rise time t_m

When a capacitor C is charged through the resistances R_A and R_B from a supply V_{CC} (or V_S) then voltage across C at any time t is given by

$$v_c = V_{CC} + Ae^{-t/C(R_A+R_B)}$$

where A is a constant determined by the initial condition.

Let at $t = 0$, $v_c = \frac{1}{3}V_{CC}$ then $A = -\frac{2}{3}V_{CC}$. Thus,

$$v_c = V_{CC} - \frac{2}{3}V_{CC}e^{-t/C(R_A+R_B)}$$

At time $t = t_m$, $v_c = \frac{2}{3}V_{CC}$. Therefore,

$$t_m = C(R_A + R_B) \ln 2$$

Calculation of discharging/fall time t_s

The capacitor discharges through R_B from $\frac{2}{3}V_{CC}$ to $\frac{1}{3}V_{CC}$ during the time t_s . The capacitor voltage during this discharging period is given by

$$v_c = \frac{2}{3}V_{CC}e^{-t/R_B C}$$

Therefore, t_s is given by

$$\frac{1}{3}V_{CC} = \frac{2}{3}V_{CC}e^{-t_s/R_B C} \Rightarrow t_s = CR_B \ln 2$$

Therefore, the **total time period** of the output waveform is

$$T = t_m + t_s = C(R_A + 2R_B) \ln 2 \simeq 0.693 \times C(R_A + 2R_B)$$

And therefore, the frequency of the resulting square-wave output is

$$f_{Th} = 1/T \simeq 1.44 \times \frac{1}{C(R_A + 2R_B)}$$

3 Data Analysis

Sl. No.	C (μ F)	R_A ($k\Omega$)	R_B ($k\Omega$)	f_{Th} (Hz)	T_m (ms) [Theory]	T_s (ms) [Theory]	T_m (ms) [Exp]	T_s (ms) [Exp]	$T = T_m + T_s$	f_{exp} (Hz)	% Error
1	0.001	1	10	66666.667	0.007623	0.00693	0.0088	0.0076	0.0164	60975.610	8.54
2	0.001	10	100	6666.667	0.07623	0.0693	0.077	0.068	0.1450	6896.552	3.45
3	0.001	100	1000	666.667	0.7623	0.693	0.76	0.68	1.4400	694.444	4.17
4	0.01	1	10	6666.667	0.07623	0.0693	0.0812	0.074	0.1552	6443.299	3.35
5	0.01	10	100	666.667	0.7623	0.693	0.86	0.77	1.6300	613.497	7.98
6	0.01	100	1000	66.667	7.623	6.93	8.7	7.7	16.4000	60.976	8.54
7	0.1	1	10	666.667	0.7623	0.693	0.62	0.52	1.14	877.193	31.58
8	0.1	10	100	66.667	7.623	6.93	6.68	6.24	12.92	77.399	16.10
9	0.1	100	1000	6.667	76.23	69.3	65.2	57.6	122.80	8.143	22.15
10	1	1	10	66.667	7.623	6.93	7.8	6.6	14.4	69.444	4.17
11	1	10	100	6.667	76.23	69.3	82	70	152.0	6.579	1.32
12	1	100	1000	0.667	762.3	693	792	696	1488.0	0.672	0.81
13	10	1	10	6.667	76.23	69.3	76	66	142	7.042	5.63
14	10	10	100	0.667	762.3	693	800	700	1500	0.667	0.00
15	10	100	1000	0.067	7623	6930	7880	6920	14800	0.068	1.35

Table 1: Comparison of Theoretical and Experimental Results for Various RC Configurations in this experiment

4 Result

The experimentally obtained results are in close agreement with the theoretically calculated ones. The mean percentage error for all the readings with respect to their respective calculated values was found to be 7.94%.

5 Conclusion

In this experiment, we learned how to use the 555 Timer IC as an astable multivibrator operating at various frequencies. We found that the measured frequency of the square waves closely matched the theoretical calculations.

6 Sources of Error

- Loose connections or improper placement of wires on the breadboard could introduce additional resistance and voltage drops.
- The output of the given voltage source might not be steady and was fluctuating.
- The Timer was assumed to be ideal, therefore additional errors due to imperfect resistance values also contributed to the errors.
- Digital oscilloscope readings may also be subjected to various errors:
 - **Sampling Rate Limitations** : If the sampling rate isn't high enough relative to the signal frequency (violating the Nyquist theorem), it can lead to aliasing and inaccurate waveform representation.
 - **Quantization Error**: Digital scopes convert analog signals to digital using an ADC (Analog-to-Digital Converter). Limited resolution can also introduce rounding or quantization noise.
 - **Timebase Accuracy**: Oscilloscopes have a timebase clock which might have slight inaccuracies, leading to timing or frequency measurement errors.
 - Temperature variations may also affect circuit behavior or measurement components.

References

- *B. Ghosh*, Fundamental Principles of Electronics 2021. ISBN: 9788187134398